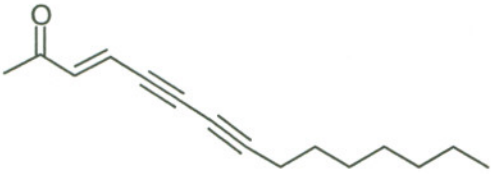
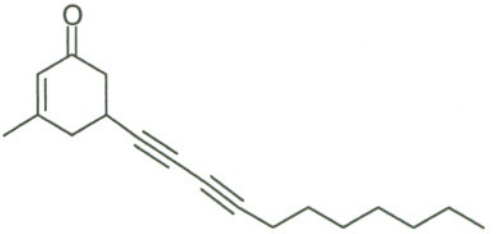
Organic Cumulative Exam February 23rd, 2001

The following diynes, designated as montiporynes, were isolated from a stony coral (*Montipora* sp.). Match the structures of montiporynes A-F with the ^{1}H NMR data sets. Briefly indicate what criteria you used to distinguish them.

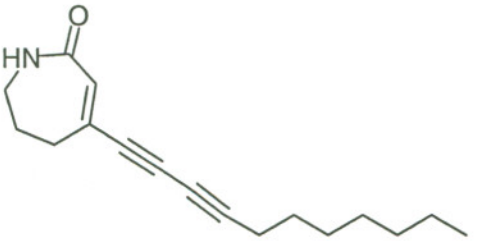
A



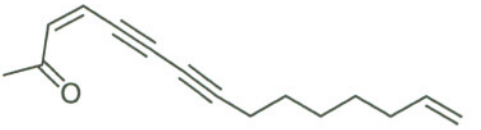
B



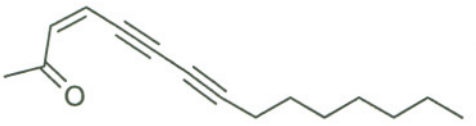
C



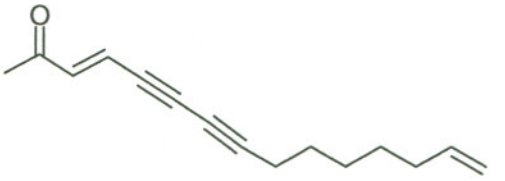
D



E



F



1

2.25 (s)
6.57 (d, 16.1)
6.73 (d, 16.1)
2.39 (t, 6.9)
1.56 (quint, 7.4)
1.42 (quint, 5.2)
1.29–1.35 (m)
0.91 (t, 7.0)

2

6.60 (s)
2.75 (td, 6.5, 2.0)
1.86 (quint, 5.4)
3.35 (t, 5.7)
2.39 (t, 6.6)
1.56 (quint, 7.4)
1.42 (quint, 7.5)
1.29–1.34 (m)
1.29–1.34 (m)
1.29–1.34 (m)
0.91 (t, 7.1)

3

2.25 (s)
6.55 (d, 16.1)
6.72 (d, 16.1)
2.38 (t, 6.9)
1.56 (quint, 7.2)
1.28–1.42 (m)
1.28–1.42 (m)
2.10 (quart, 6.8)
5.80 (ddt, 17.0, 10.2, 6.8)
4.91 (dd, 10.2, 2.1)
4.98 (dd, 17.0, 2.1)

4

2.38 (s)
6.28 (d, 11.5)
6.46 (d, 11.5)
2.39 (t, 6.9)
1.56 (quint, 7.3)
1.29–1.42 (m)
1.29–1.42 (m)
0.90 (t, 7.0)

5

5.89 (s)
2.62 (dd, 16.7, 4.6)
2.49 (dd, 16.7, 9.0)
3.19 (tt, 9.0, 4.6)
2.57 (dd, 16.4, 4.6)
2.42 (dd, 16.4, 9.0)
2.00 (s)
2.25 (t, 6.6)
1.50 (quint, 7.3)
1.39 (quint, 6.9)
1.29–1.35 (m)
1.29–1.35 (m)
1.29–1.35 (m)
0.90 (t, 7.0)

6

2.38 (s)
6.28 (d, 11.5)
6.46 (d, 11.5)
2.38 (t, 6.9)
1.56 (quint, 7.2)
1.28–1.42 (m)
1.28–1.42 (m)
2.10 (quart, 6.8)
5.80 (ddt, 17.0, 10.2, 6.8)
4.91 (dd, 10.2, 2.1)
4.98 (dd, 17.0, 2.1)